CHEMISTRY

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

51. For a $Mg|Mg^{2+}(aq) \,||\, Ag^{+}(aq) \,|\, Ag$ the correct Nernst equation is :

(1) $E_{cell} = E^{\circ}_{cell} + \frac{RT}{2F}\ln\frac{[Ag^{+}]^2}{[Mg^{2+}]}$

(2) $E_{cell} = E^{\circ}_{cell} - \frac{RT}{2F}\ln\frac{[Ag^{+}]^2}{[Mg^{2+}]}$

(3) $E_{cell} = E^{\circ}_{cell} - \frac{RT}{2F}\ln\frac{[Ag^{+}]}{[Mg^{2+}]}$

(4) $E_{cell} = E^{\circ}_{cell} - \frac{RT}{2F}\ln\frac{[Mg^{2+}]}{[Ag^{+}]}$

Answer (1)

Sol.

Cathode	$(Ag^{+}(aq) + e^{-} \rightarrow Ag) \times 2$	
Anode	Mg	$\rightarrow Mg^{2+}(aq) + 2e^{-}$
Cell reaction	$2Ag^{+} + Mg$	$\rightarrow 2Ag + Mg^{2+}$

$$Q = \frac{[Mg^{2+}]}{[Ag^{+}]^2}$$

By Nernst equation

$$E_{cell} = E^{o}_{cell} - \frac{RT}{nF}\ln Q$$

$$E_{cell} = E^{o}_{cell} - \frac{RT}{nF}\ln\frac{[Mg^{2+}]}{[Ag^{+}]^2}$$

$$= E^{o}_{cell} + \frac{RT}{2F}\ln\frac{[Ag^{+}]^2}{[Mg^{2+}]}$$

52. 1.24 g of AX_2 (molar mass 124 g mol^{-1}) is dissolved in 1 kg of water to form a solution with boiling point of 100.0156°C, while 25.4 g of AY_2 (molar mass 250 g mol^{-1}) in 2 kg of water constitutes a solution with a boiling point of 100.0260°C.

$K_b(H_2O)$ = 0.52 K kg mol^{-1}

Which of the following is **correct** ?

(1) AX_2 and AY_2 (both) are fully ionised

(2) AX_2 is fully ionised while AY_2 is completely unionised

(3) AX_2 and AY_2 (both) are completely unionised

(4) AX_2 is completely unionised while AY_2 is fully ionised

Answer (2)

Sol. For AX_2

$\Delta T_b = iK_bm$

$$0.0156 = i \times 0.52 \times \frac{1.24}{124 \times 1}$$

$3 = i$

$3 = 1 + 2\alpha$

$1 = \alpha$

For AY_2

$\Delta T_b = iK_bm$

$$0.0260 = i \times 0.52 \times \frac{25.4}{250 \times 2}$$

$i \approx 1$

$\therefore$ AX_2 is completely ionised & AY_2 is completely unionised



53. The standard reduction potential values of some of the p-block ions are given below. Predict the one with the strongest oxidising capacity.

(1) $E^{\ominus}_{Al^{3+}/Al} = -1.66$ V

(2) $E^{\ominus}_{Sn^{4+}/Sn^{2+}} = +1.15$ V

(3) $E^{\ominus}_{Tl^{3+}/Tl} = +1.26$ V

(4) $E^{\ominus}_{Pb^{4+}/Pb^{2+}} = +1.67$ V

Answer (4)

Sol. The element having strongest oxidising capacity will have highest value of standard reduction potential

54. The reaction $A_2 + B_2 \rightarrow 2AB$ follows the mechanism

$$A_2 \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} A + A \text{ (fast)}$$

$$A + B_2 \xrightarrow{k_2} AB + B \text{ (slow)}$$

$A + B \rightarrow AB$ (fast)

The overall order of the reaction is :

(1) 1.5 (2) 2.5

(3) 3 (4) 2

Answer (1)

Sol. Since, second step is slow step

$r = k_2[A][B_2]$ ...(i)

Also, $\frac{k_1}{k_{-1}} = \frac{[A]^2}{[A_2]}$

$$[A] = \left(\frac{[A_2]k_1}{k_{-1}}\right)^{\frac{1}{2}} \quad \text{...(ii)}$$

$$r = k_2\left(\frac{k_1}{k_{-1}}\right)^{\frac{1}{2}}[A_2]^{\frac{1}{2}}[B_2]$$

$$\text{order} = \frac{1}{2} + 1 = \frac{3}{2}$$

55. The molar conductivity of a weak electrolyte when plotted against the square root of its concentration, which of the following is expected to be observed?

(1) A small increase in molar conductivity is observed at infinite dilution

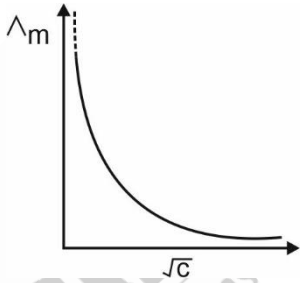
(2) Molar conductivity increases sharply with increase in concentration

(3) A small decrease in molar conductivity is observed at infinite dilution

(4) Molar conductivity decreases sharply with increase in concentration.

Answer (4)

Sol. For weak electrolyte, variation of $\wedge_m$ with $\sqrt{c}$ is



56. At temperature T, compound $AB_2(g)$ dissociates as $AB_2(g) \rightleftharpoons AB(g) + \frac{1}{2}B_2(g)$ having degree of dissociation x (small compared to unity). The correct expression for x in terms of K_p and p is

(1) $\sqrt[3]{\frac{2K_p^2}{p}}$ (2) $\sqrt[3]{\frac{2K_p}{p}}$

(3) $\sqrt[4]{\frac{2K_p}{p}}$ (4) $\sqrt{K_p}$

Answer (1)

Sol.

	$AB_2(g)$	$\rightleftharpoons$	$AB(g)$	+	$\frac{1}{2}B_2(g)$
$t = 0$	p_0				
$t = t_{eq}$	$p_0(1-x)$		p_0x		$\frac{p_0x}{2}$

$$p = p_0x - p_0x + p_0x + \frac{p_0x}{2}$$



$$p = p_0\left(1+\frac{x}{2}\right)$$

$$p_0 = \frac{p}{\left(1+\frac{x}{2}\right)}$$

$$K_p = \frac{(p_{AB})(p_{B_2})^{1/2}}{(p_{AB_2})}$$

$$K_p = \frac{(p_0 x)\left(\frac{p_0 x}{2}\right)^{1/2}}{p_0(1-x)}$$

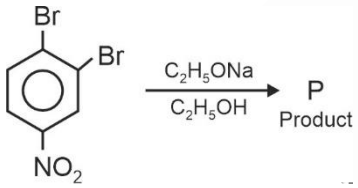
$$K_p = \frac{\frac{px}{\left(1+\frac{x}{2}\right)}\left(\frac{p}{1+\frac{x}{2}}\times\frac{x}{2}\right)^{1/2}}{\frac{p(1-x)}{\left(1+\frac{x}{2}\right)}}$$

Since $x << 1$

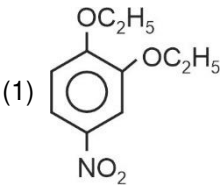
$$K_p = \frac{p^{1/2}x^{3/2}}{2^{1/2}}$$

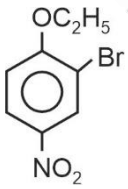
$$x = \sqrt[3]{\frac{2K_p^2}{p}}$$

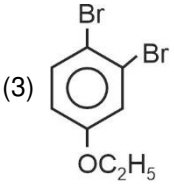
57. In the following substitution reaction :

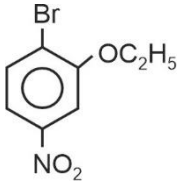


Product 'P' formed is

(1) 

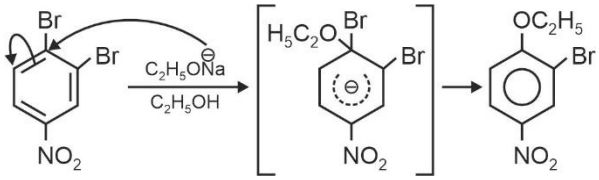
(2) 

(3) 

(4) 

Answer (2)

Sol. Br at the para of NO_2 will undergo aromatic nucleophilic substitution by nucleophile C_2H_5ONa.



58. Total number of nucleophiles from the following is

NH_3, PhSH, $(H_3C)_2S$, $H_2C{=}CH_2$, $\bar{O}H$, H_3O^+, $(CH_3)_2CO$, $\rangle{=}NCH_3$

(1) 4 (2) 7

(3) 5 (4) 6

Answer (3)

Sol. Any specie having electrons available for donation can act as nucleophile.

Total 5 are present.

59. Choose the **correct** statements :

(A) Weight of a substance is the amount of matter present in it.

(B) Mass is the force exerted by gravity on an object.

(C) Volume is the amount of space occupied by a substance.

(D) Temperatures below 0°C are possible in Celsius scale, but in Kelvin scale negative temperature is not possible.

(E) Precision refers to the closeness of various measurements for the same quantity.

Choose the **correct** answer from the options given below.

(1) (A), (B) and (C) only (2) (A), (D) and (E) only

(3) (C), (D) and (E) only (4) (B), (C) and (D) only



Answer (3)

Sol. Mass of substance is amount of matter present in it. Weight is force exerted by gravity on object.

60. Match **List-I** with **List-II**.

List-I (Carbohydrate)	**List-II (Linkage Source)**
(A) Amylose	(I) β-C_1-C_4, plant
(B) Cellulose	(II) α-C_1-C_4, animal
(C) Glycogen	(III) α-C_1-C_4, α-C_1-C_6, plant
(D) Amylopectin	(IV) α-C_1-C_4, plant

Choose the **correct** answer from the options given below.

(1) (A)-(IV), (B)-(I), (C)-(III), (D)-(II)

(2) (A)-(II), (B)-(III), (C)-(I), (D)-(IV)

(3) (A)-(III), (B)-(II), (C)-(I), (D)-(IV)

(4) (A)-(IV), (B)-(I), (C)-(II), (D)-(III)

Answer (4)

Sol. Amylose $\Rightarrow$ It is a plant based starch it has α-C_1-C_4 glycosidic linkage.

Cellulose $\Rightarrow$ It has β-C_1-C_4 glycosidic linkage

Glycogen $\Rightarrow$ It has α-C_1-C_4 and glycosidic linkage (animal starch)

Amylopectin $\Rightarrow$ It is a plant based with α-C_1-C_4 and C_1-C_6 glycosidic linkage

61. If a_0 is denoted as the Bohr radius of hydrogen atom, then what is the de-Broglie wavelength (λ) of the electron present in the second orbit of hydrogen atom? [n : any integer]

(1) $\frac{2a_0}{n\pi}$ (2) $\frac{4n}{\pi a_0}$

(3) $\frac{4\pi a_0}{n}$ (4) $\frac{8\pi a_0}{n}$

Answer (4)

Sol. $r_n = \frac{a_0 n^2}{z}$

Also, $2\pi r_n = n\lambda$

Where λ is de-Broglie wavelength

$$\frac{2\pi a_0 n^2}{z} = n\lambda$$

For second orbit of H-atom

$$\lambda = \frac{8\pi a_0}{n}$$

62. Match **List - I** with **List - II.**

List - I (Complex)	**List - II (Hybridisation & Magnetic characters)**
(A) $[MnBr_4]^{2-}$	(I) d^2sp^3 & diamagnetic
(B) $[FeF_6]^{3-}$	(II) sp^3d^2 & paramagnetic
(C) $[Co(C_2O_4)_3]^{3-}$	(III) sp^3 & diamagnetic
(D) $[Ni(CO)_4]$	(IV) sp^3 & paramagnetic

Choose the **correct** answer from the options given below :

(1) (A)-(IV), (B)-(II), (C)-(I), (D)-(III)

(2) (A)-(III), (B)-(I), (C)-(II), (D)-(IV)

(3) (A)-(IV), (B)-(I), (C)-(II), (D)-(III)

(4) (A)-(III), (B)-(II), (C)-(I), (D)-(IV)

Answer (1)

Sol. $[MnBr_4]^{2-}$

Mn^{2+}, Br^- is WFL

d^5, so it is sp^3 and paramagnetic.

$[FeF_6]^{3-}$

Fe^{3+}, F^- is WFL

d^5, so it is sp^3d^2 and paramagnetic.



$[Co(C_2O_4)_3]^{3-}$

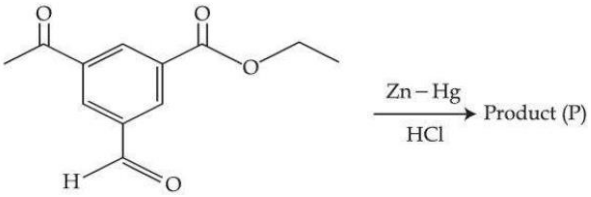
Co^{3+}, $C_2O_4^{2-}$ is a SFL

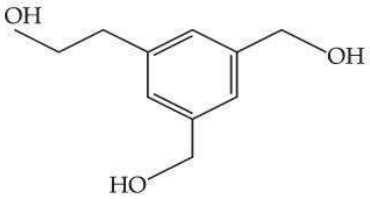
d^6, so it is d^2sp^3 and diamagnetic.

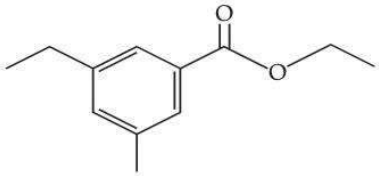
$[Ni(CO)_4]$, CO is a SFL

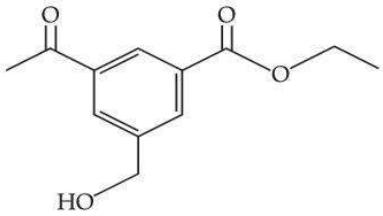
d^{10}, it is sp^3 and diamagnetic.

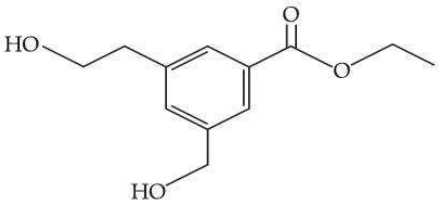
63. The product (P) formed in the following reaction is:



(1) 

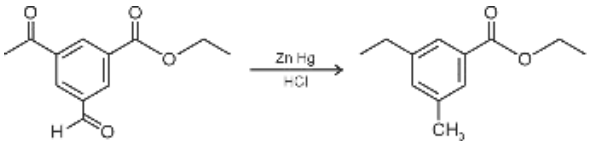
(2) 

(3) 

(4) 

Answer (2)

Sol. It is Clemmensen reduction, it will not reduce ester,



Ester cannot be reduced by clemmensen reduction

64. The correct increasing order of stability of the complexes based on Δ_0 value is:

I. $[Mn(CN)_6]^{3-}$ II. $[Co(CN)_6]^{4-}$

III. $[Fe(CN)_6]^{4-}$ IV. $[Fe(CN)_6]^{3-}$

(1) III < II < IV < I (2) II < III < I < IV

(3) IV < III < II < I (4) I < II < IV < III

Answer (4)

Sol. Neglecting pairing energy

I. $[Mn(CN)_6]^{3-} \Rightarrow Mn^{3+}$, t_{2g}^4, CFSE = $-0.4 \times 4\,\Delta_o = -1.6\Delta_o$

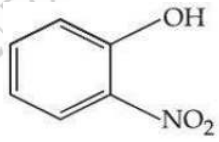
II. $[Co(CN)_6]^{4-} \Rightarrow Co^{2+}$, $t_{2g}^6 e_g^1$, CFSE = $-0.4 \times 6 + 0.6 \times 1 = -1.8\Delta_o$

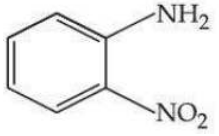
III. $[Fe(CN)_6]^{4-} \Rightarrow Fe^{2+}$, $t_{2g}^6 e_g^0$, CFSE = $-0.4 \times 6 = -2.4\Delta_o$

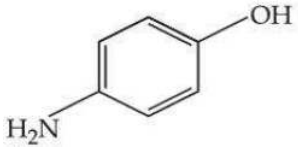
IV. $[Fe(CN)_6]^{3-} \Rightarrow Fe^{3+}$, $t_{2g}^5 e_g^0$, CFSE = $-0.4 \times 5 = -2\Delta_o$

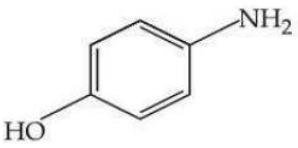
Order of stability III > IV > II > I

65. The steam volatile compounds among the following are :

(A) 

(B) 

(C) 

(D) 



Choose the **correct** answer from the options given below :

(1) (B) and (D) Only

(2) (A) and (C) Only

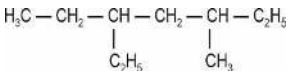
(3) (A), (B) and (C) Only

(4) (A) and (B) Only

Answer (4)

Sol. Ortho nitro phenol and ortho nitro aniline will be steam volatile as they will show intra molecular H-bonding.

66. Match List I with List-II

	List-I (Structure)	**List-II (IUPAC Name)**
(A)		(I) 4-Methylpent-l-ene
(B)	$(CH_3)_2C(C_3H_7)_2$	(II) 3-Ethyl-5-methylheptane
(C)		(III) 4, 4-Dimethylheptane
(D)		(IV) 2-Methyl-1, 3-pentadiene

Choose the correct answer from the options given below

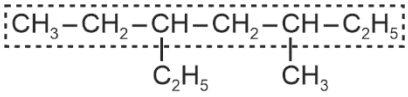
(1) (A)-(II), (B)-(III), (C)-(I), (D)-(IV)

(2) (A)-(II), (B)-(III), (C)-(IV), (D)-(I)

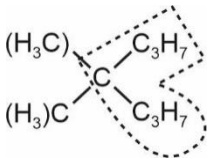
(3) (A)-(III), (B)-(II), (C)-(IV), (D)-(I)

(4) (A)-(III), (B)-(II), (C)-(I), (D)-(IV)

Answer (2)

Sol. (A) 

3-Ethyl-5-methylheptane

(B) 

4,4-Dimethylheptane

(C) 

2-Methyl-1,3-pentadiene

(D) 

4-Methylpent-1-ene

67. 500 J of energy is transferred as heat to 0.5 mol of Argon gas at 298 K and 1.00 atm. The final temperature and the change in internal energy respectively are: Given: R = 8.3 JK^{-1} mol^{-1}

(1) 378 K and 500 J (2) 368 K and 500 J

(3) 378 K and 300 J (4) 348 K and 300 J

Answer (4)

Sol. At cons. P

$Q = nC_p \Delta T$

$$500 = 0.5 \times \frac{5}{2} \times 8.3 \left(T_f - 298\right)$$

$346.2\ K = T_f$

$\Delta U = nC_v\Delta T$

$$= \frac{1}{2} \times \frac{3}{2} \times 8.3 \times \left(346.2 - 298\right)$$

$= 300\ J$

68. The correct option with order of melting points of the pairs (Mn, Fe), (Tc, Ru) and (Re, Os) is:

(1) Fe < Mn, Ru < Tc and Re < Os

(2) Mn < Fe, Tc < Ru and Os < Re

(3) Fe < Mn, Ru < Tc and Os < Re

(4) Mn < Fe, Tc < Ru and Re < Os

Answer (2)

Sol. Melting point order

Fe > Mn

Ru > Tc

Re > Os



69. An element 'E' has the ionisation enthalpy value of 374 kJ mol^{-1}.'E' reacts with elements A, B, C and D with electron gain enthalpy values of –328, –349, –325 and –295 kJ mol^{-1}, respectively. The correct order of the products EA, EB, EC and ED in terms of ionic character is:

(1) ED > EC > EA > EB (2) EA > EB > EC > ED

(3) EB > EA > EC > ED (4) ED > EC > EB > EA

Answer (3)

Sol. The element having high value of Electron gain enthalpy (magnitude) will form a compound having higher ionic character so order of ionic character

EB > EA > EC > ED

70. Given below are two statements :

Statement (I) : The radii of isoelectronic species increases in the order.

$Mg^{2+} < Na^{+} < F^{-} < O^{2-}$

Statement (II) : The magnitude of electron gain enthalpy of halogen decreases in the order.

Cl > F > Br > I

In the light of the above statements, choose the **most appropriate answer** from the options given below:

(1) **Statement I** is correct but **Statement II** is incorrect

(2) Both **Statement I** and **Statement II** are correct

(3) **Statement I** is incorrect but **Statement II** is **correct**

(4) Both **Statement I** and **Statement II** are incorrect

Answer (2)

Sol. $r \propto q^{-}$ (for isoelectronic species)

$$\propto \frac{1}{q^{+}}$$

$\therefore$ Statement I is correct

Magnitude of electron gain enthalpy

Cl > F > Br > I

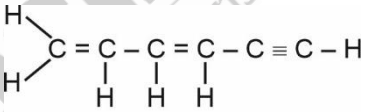
SECTION - B

Numerical Value Type Questions: This section contains 5 Numerical based questions. The answer to each question should be rounded-off to the nearest integer.

71. The sum of sigma (σ) and pi(π) bonds in Hex-1, 3-dien-5-yne is _____.

Answer (15)

Sol.



Hex-1, 3-dien-5-yne

No. of π bond = 4

No. of σ bond = 11

72. If A_2B is 30% ionised in an aqueous solution, then the value of van't Hoff factor (i) is _____ × 10^{-1}.

Answer (16)

Sol.

AB_2	$\rightleftharpoons$	A^{2+}	+	$2B^{-}$
1		0		0
$1-\alpha$		α		2α

$i = 1 + 2\alpha$

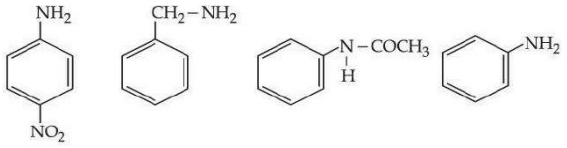
$= 1 + 2 \times (0.3)$

$= 1.6$

$= 16 \times 10^{-1}$



73. Given below are some nitrogen containing compounds

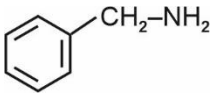


Each of them is treated with HCl separately, 1.0 g of the most basic compound will consume _____ mg of HCl.

(Given molar mass in g mol^{-1} C : 12, H : 1, O : 16, Cl : 35.5)

Answer (341)

Sol. The most basic compound will be aliphatic amine due to localised electrons



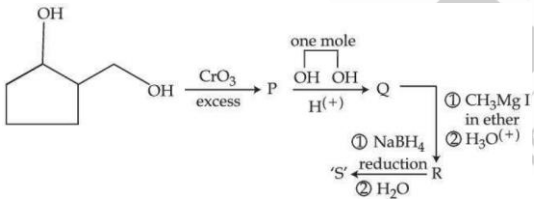
1 mole of this will consume 1 mole HCl

So mass of HCl consumed for 1 g of this compound

$$= \frac{1}{107} \times 36.5$$

= 0.341 gm

= 341 mg

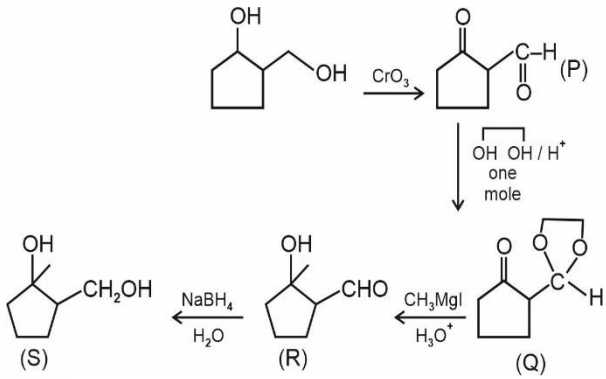
74. 

0.1 mole of compound 'S' will weight _____ g.

(Given molar mass in g mol^{-1} C : 12, H : 1, O : 16)

Answer (13)

Sol.



mass of 0.1 mole (S) = 0.1(84 + 32 +14)

= 13 g

75. The molar mass of the water insoluble product formed from the fusion of chromite ore ($FeCr_2O_4$), with Na_2CO_3 in presence of O_2 is _____ g mol^{-1}.

Answer (160)

Sol. $FeCr_2O_4 + Na_2CO_3 + O_2 \rightarrow Na_2CrO_4 + Fe_2O_3 + CO_2$

Insoluble product will be Fe_2O_3

molar mass = 56 × 2 + 16 × 3

= 112 + 48

= 160

❐ ❐ ❐

